

BENSON, MN, USA

WMO: 727507

Lat: 45.332N

Lon: 95.651W

Elev: 317

StdP: 97.58

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 54904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.3	(c) -24.0	(d) -31.1	(e) 0.2	(f) -25.8	(g) -27.7	(h) 0.3	(i) -23.1	(j) 14.3	(k) -11.5	(l) 12.8	(m) -13.6	(n) 2.6	(o) 350	(p) 0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	32.4	22.5	31.0	22.1	28.9	21.0	25.0	29.7	23.8	28.3	22.7	27.2	5.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.2	28.0	22.4	17.7	26.9	21.2	16.5	25.9	78.0	29.8	72.8	28.2	68.3	27.3	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.8	(b) 10.4	(c) 8.9	(e) -30.6	(f) 35.1	(g) 3.7	(h) 1.9	(i) -33.2	(j) 36.5	(k) -35.4	(l) 37.6	(m) -37.4	(n) 38.7	(o) -40.1	(p) 40.0		
			DB													
			WB													
				-30.6	26.8	3.6	1.3	-33.2	27.7	-35.3	28.4	-37.4	29.1	-40.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-10.7	-8.9	-1.3	6.9	13.9	19.5	21.9	20.2	16.2	8.5	0.0	-7.6		
	DBStd	12.85	7.61	7.54	7.55	5.83	4.98	3.85	3.36	3.28	4.92	5.23	6.29	7.00		
	HDD10.0	2617	642	528	355	126	19	0	0	0	7	92	304	544		
	HDD18.3	4588	900	762	609	346	154	31	8	19	99	308	551	803		
	CDD10.0	1387	0	0	5	32	141	286	368	316	192	45	3	0		
	CDD18.3	315	0	0	0	1	17	67	118	76	34	3	0	0		
	CDH23.3	3306	0	0	1	31	243	692	1210	731	352	45	1	0		
	CDH26.7	1048	0	0	0	9	80	218	418	205	107	11	0	0		
(14) Wind	WSAvg	3.8	4.1	4.2	4.3	4.5	4.3	3.6	3.0	2.8	3.4	4.0	4.1	3.9		
(15) Precipitation	PrecAvg	684	19	17	38	65	75	103	99	89	77	64	28	18		
	PrecMax	921	46	39	97	180	149	183	194	167	152	170	88	51		
	PrecMin	484	2	4	12	22	10	40	18	25	25	8	1	5		
	PrecStd	108	13	10	22	39	32	37	45	41	38	45	24	13		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	12.0	21.2	27.3	32.6	33.1	34.8	32.9	32.4	27.8	19.8	9.7		
		MCWB	3.3	7.9	12.7	14.6	18.7	22.5	23.8	24.1	20.9	16.6	11.7	4.4		
	2%	DB	3.0	6.3	16.2	22.8	28.7	31.3	32.4	31.0	29.1	23.0	15.9	6.2		
		MCWB	1.0	3.4	10.6	11.8	17.5	21.3	23.8	23.2	20.4	14.2	9.4	2.9		
	5%	DB	1.8	3.0	12.4	20.0	26.2	28.9	31.0	28.8	27.5	20.0	12.1	2.9		
		MCWB	0.5	1.3	8.6	11.0	15.8	19.9	23.2	21.9	19.4	14.0	7.8	1.2		
	10%	DB	0.0	2.1	8.7	17.2	22.9	27.4	28.8	27.4	25.0	17.3	8.7	2.1		
		MCWB	-0.8	0.8	6.1	9.7	14.4	19.0	22.0	20.8	18.1	11.8	5.5	0.7		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	7.1	15.0	16.1	21.6	25.0	27.1	26.0	24.0	19.0	13.7	6.2		
		MCDB	6.4	10.3	17.8	23.9	28.2	30.6	31.4	29.5	28.9	23.7	18.1	8.5		
	2%	WB	1.4	3.8	11.5	13.9	19.3	23.2	25.5	24.5	22.5	16.6	10.1	3.2		
		MCDB	3.1	6.1	15.1	20.2	25.5	28.3	30.4	28.7	27.2	21.5	14.5	5.5		
	5%	WB	0.4	1.9	8.5	12.1	17.6	21.8	24.2	23.3	21.0	14.4	7.8	1.6		
		MCDB	1.5	3.3	12.1	17.9	23.0	27.5	28.7	27.4	25.3	18.8	11.0	3.2		
	10%	WB	-1.1	0.5	5.8	10.4	15.9	20.5	23.0	22.0	19.5	12.4	5.6	0.3		
		MCDB	-0.3	1.8	8.8	15.9	21.5	25.4	27.5	26.0	23.8	16.6	8.8	1.7		
(35) Mean Daily Temperature Range	MDBR		10.1	10.4	10.2	12.1	12.6	11.9	12.3	12.5	13.2	12.0	10.3	9.2		
	5% DB	MCDBR	11.2	10.9	15.0	17.8	16.8	14.2	13.5	13.7	15.6	16.7	15.4	11.7		
		MCWBR	9.6	8.8	9.7	9.0	8.0	6.6	6.7	7.2	7.7	9.1	9.8	9.2		
	5% WB	MCDBR	10.4	10.3	14.2	15.0	13.6	11.9	11.7	11.9	12.8	13.5	14.2	10.9		
		MCWBR	9.1	8.2	9.8	8.9	7.6	6.7	6.8	7.3	7.7	8.5	9.6	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.278	0.308	0.363	0.388	0.394	0.394	0.403	0.365	0.316	0.281	0.258		
	taud		2.436	2.401	2.431	2.329	2.319	2.360	2.348	2.310	2.398	2.525	2.533	2.468		
	Ebn at Noon		863	914	929	896	881	874	869	846	853	860	840	835		
	Edh at Noon		69	87	98	119	124	120	120	120	101	77	63	60		
(44) All-Sky Solar Radiation	RadAvg		1.75	2.83	3.80	4.73	5.43	6.05	6.48	5.39	4.14	2.63	1.70	1.24		
	RadStd		0.21	0.32	0.35	0.41	0.41	0.40	0.23	0.45	0.31	0.37	0.16	0.15		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page